

A Thomson–Brennan Theorem for Balanced Finite Covers

A partial solution to the Open Problem of arXiv:2112.08504

Lane 19, agent_lane_19, model GPT5.6

Status. candidate_partial_solution_likely_valid.

Source problem. Biswas–Putinar ask on PDF page 6 whether, for every atomless positive measure compactly supported in a relatively compact finite-connectivity domain U of a finite-genus open Riemann surface, $\mathcal{O}(U)$ is dense in $L^2(\mu)$ exactly when there are no corresponding analytic bounded point evaluations [1].

Open Problem. *Let X be an open Riemann surface of finite genus, and let μ be a positive Borel measure on X without point masses. Let $U \subset X$ be an open, relatively compact subset of X , with finitely many components of $X \setminus U$, none reduced to a point. Assume the closed support of the measure μ is contained in U . Then analytic functions $\mathcal{O}(U)$ are dense in $L^2(\mu)$ if and only if there are no corresponding bounded analytic point evaluations.*

Definitions. Let $\Omega \subset \mathbb{C}$ be a domain and let $\pi : U \rightarrow \Omega$ be a finite unbranched holomorphic covering of degree d . For an atomless finite positive Borel measure ν with compact support $K \Subset \Omega$, define its *balanced lift* μ by

$$\int_U F d\mu = \frac{1}{d} \int_{\Omega} \sum_{w \in \pi^{-1}(z)} F(w) d\nu(z) \quad (F \geq 0 \text{ Borel}). \quad (1)$$

Write $H^2(\Omega, \nu) = \overline{\mathcal{O}(\Omega)}^{L^2(\nu)}$ and similarly on U . A point is an analytic bounded point evaluation (ABPE) if evaluation is bounded with a locally uniform bound throughout a neighborhood.

Proof intuition. A function on the cover is a holomorphic section of the direct-image bundle $E = \pi_* \mathcal{O}_U$, whose fiber consists of functions on the d -point fiber of π . The bundle is holomorphically trivial because the base is an open Riemann surface. The balanced lift equips it with the ordinary counting metric. In any global holomorphic frame that metric is uniformly equivalent, on K , to the Euclidean metric. Thus mean approximation upstairs is exactly d independent copies of mean approximation downstairs, up to an equivalent norm. Point evaluations pass through the same frame, so Brennan’s planar dichotomy completes the argument.

Theorem 1 (balanced finite-cover case). *Let $\Omega \subset \mathbb{C}$ have finitely many complementary components, each of positive diameter. Let $\pi : U \rightarrow \Omega$ be a finite unbranched holomorphic covering, let ν be atomless and compactly supported in Ω , and let μ be its balanced lift (1). Then:*

$$\begin{aligned} H^2(U, \mu) = L^2(\mu) &\iff H^2(\Omega, \nu) = L^2(\nu), \\ H^2(U, \mu) \text{ has ABPEs} &\iff H^2(\Omega, \nu) \text{ has ABPEs.} \end{aligned}$$

Consequently $\mathcal{O}(U)$ is dense in $L^2(\mu)$ if and only if it has no ABPEs.

Proof. The direct image $E = \pi_* \mathcal{O}_U$ is a rank- d holomorphic vector bundle on Ω : over an evenly covered disk, a section is simply a d -tuple of holomorphic functions. Every holomorphic vector bundle on a

noncompact Riemann surface is holomorphically trivial [3, Section 30, in particular Theorem 30.4]. Choose a global holomorphic frame $s_1, \dots, s_d$. Hence every $g \in \mathcal{O}(U)$ has a unique representation

$$g = \sum_{j=1}^d (a_j \circ \pi) s_j, \quad a = (a_1, \dots, a_d) \in \mathcal{O}(\Omega)^d. \quad (2)$$

Give E_z the normalized counting metric

$$\langle v, u \rangle_z = \frac{1}{d} \sum_{w \in \pi^{-1}(z)} v(w) \overline{u(w)}.$$

Let $G(z) = (\langle s_j, s_k \rangle_z)_{j,k}$. It is a continuous positive definite matrix. Compactness of $K = \text{supp } \nu$ gives constants $0 < c \leq C < \infty$ such that

$$c \|\xi\|_2^2 \leq \xi^* G(z) \xi \leq C \|\xi\|_2^2 \quad (z \in K, \xi \in \mathbb{C}^d). \quad (3)$$

Equations (1) and (2) yield

$$\|g\|_{L^2(\mu)}^2 = \int_K a(z)^* G(z) a(z) d\nu(z). \quad (4)$$

The same fiberwise correspondence identifies all of $L^2(\mu)$ with the measurable L^2 sections of E . In the chosen frame, (3) says that its norm is equivalent to the standard norm on $L^2(\nu)^d$. Under this identification $\mathcal{O}(U)$ is exactly $\mathcal{O}(\Omega)^d$. Therefore it is dense if and only if $\mathcal{O}(\Omega)$ is dense in $L^2(\nu)$, proving the first equivalence.

Suppose next that z is an ABPE downstairs. Each coefficient in (2) satisfies $|a_j(z)| \leq M_z \|a_j\|_{L^2(\nu)}$, with M_z locally bounded. The lower bound in (3) controls every coefficient norm by $c^{-1/2} \|g\|_{L^2(\mu)}$. Since the frame sections are locally bounded, for every $w \in \pi^{-1}(z)$,

$$|g(w)| \leq \sum_j |s_j(w)| |a_j(z)| \leq C_w \|g\|_{L^2(\mu)},$$

with locally bounded constants. Thus all points over an ABPE neighborhood are ABPEs upstairs.

Conversely, if w is an ABPE upstairs and $z = \pi(w)$, apply its estimate to $g = a \circ \pi$. Formula (1) gives $\|a \circ \pi\|_{L^2(\mu)} = \|a\|_{L^2(\nu)}$ and $g(w) = a(z)$. Hence z is an ABPE downstairs. This proves the second equivalence.

Finally, Runge approximation identifies the $L^2(\nu)$ -closure of $\mathcal{O}(\Omega)$ with the rational closure used in Brennan's theorem. Under the stated finite-connectivity and positive-diameter hypotheses, Brennan's theorem says that this closure is proper exactly when it has ABPEs [2, 1]. Combining that dichotomy with the two equivalences proves the last assertion. $\square$

Remark 1 (genuinely positive genus). *The theorem is not confined to coordinate charts. A connected double cover of a four-connected planar domain with nontrivial monodromy around each of its four boundary components is an unbranched bordered surface with four boundary components and genus one: multiplicativity of Euler characteristic gives $2 - 2g - 4 = 2(2 - 4) = -4$, hence $g = 1$. Such a cover can be placed relatively compactly in a slightly larger cover, fitting the topology of the source problem.*

Scope and remaining obstruction. The proof uses the balanced fiber law essentially. An arbitrary measure disintegrates into a possibly degenerate matrix weight in a frame of $\pi_* \mathcal{O}_U$; then (3) can fail in the only direction needed to recover coefficient evaluations. The theorem also does not treat branching that accumulates at the support. The full source problem remains open.

Novelty and verification. Searches through 2026-08-13 used the exact problem wording and combinations of “Thomson/Brennan theorem”, “finite covering”, “balanced measure”, “vector bundle”, and “matrix-valued measure”. They found no matching statement. The identification $\mathcal{O}(U) = \Gamma(\Omega, \pi_*\mathcal{O}_U)$, bundle triviality, compact Gram bounds, both BPE implications, and the Euler-characteristic example were checked independently in the accompanying verification report. Novelty confidence is moderate pending specialist review.

Human-review recommendation. Check first that the global frame of $\pi_*\mathcal{O}_U$ induces the claimed measurable L^2 identification, then the uniform lower Gram bound on K . Those two points make the reduction to Brennan exact.

References

- [1] S. Biswas and M. Putinar, *Approximation in the mean on rational curves*, arXiv:2112.08504 (2021).
- [2] J. E. Brennan, *The structure of certain spaces of analytic functions*, Comput. Methods Funct. Theory 8 (2008), 625–640.
- [3] O. Forster, *Lectures on Riemann Surfaces*, Graduate Texts in Mathematics 81, Springer, corrected printing, 1999.